# Assignment 2: Marketing Analytics Toolkit

Module Name: Marketing Analytics

Module Code: BEMM463

Candidate Number: 058931

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University of Exeter Business School

| Assignment Cover Sheet | |
|---|---|
| Candidate Number | **058931** |
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| Module Name | Marketing Analytics |
| Assignment Title | Assignment 2: Marketing Analytics Toolkit |

Within the Business School we support the responsible and ethical use of GenAI tools, and we seek to develop your ability to use these tools to help you study and learn. An important part of this process is being transparent about how you have used GenAI tools during the preparation of your assignments. Information about GenAI can be found here and guidance on the responsible use of GenAI tools can be found here.

The below declaration is intended to guide transparency in the use of GenAI tools, and to assist you in ensuring the appropriate citation of those tools within your work.

I have used GenAI tools in the production of this work.

The following GenAI tools have been used: [please specify] Gemini......

☐ I have used GenAI tools to assist with research or gathering information.

☒ I have used GenAI tools to help me understand key theories and concepts.

☒ I have used GenAI tools to help me analyse data.

☐ I have used GenAI tools to create code.

☒ I have used GenAI tools to suggest a plan or structure of my assessment.

☐ I have used GenAI tools to give me feedback on a draft.

☐ I have used GenAI tool to generate images, figures or diagrams.

☐ I have used GenAI tools to generate creative content for my work.

☒ I have used GenAI tools to proofread and correct grammar or spelling errors.

☐ Other [please specify] ..................................................................................................................................

☒ I declare that I have referenced use of GenAI tools and outputs within my assessment in line with the University guidelines for referencing GenAI in academic work.

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# MARKETING ANALYTICS TOOLKIT

## CLUSTER ANALYSIS

### What it is?

Cluster Analysis is a form of unsupervised classification that embraces the core marketing ideology that all customers are different; in fact, it classifies customers based on their similarities without predefined categories and it groups similar observations (clusters).

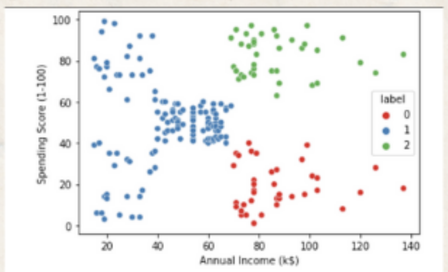
This statistical data mining technique is crucial for identifying patterns and relationships within diverse datasets, aiming to create homogeneous segments from a heterogeneous whole.

### how it works?

- To tackle the challenge of heterogeneity among customers, cluster analysis processes a dataset to form groups based on similarities that exist within the data.
- Common methods: K-Means, Hierarchical Clustering.

Euclidean Distance: This formula calculates the straight-line distance between two points

$$d(x, y) = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2}$$




*The graph segments mall customers into three groups based on income and spending score. "Savers" (red) have high income, low spending; "Carefree" (blue) have low income, but spend; and "Spenders" (green) have high income and high spending.*

Source: Neptune.ai (n.d.). K-Means Clustering. neptune.ai.

### Managerial Implications

- Using cluster analytics will help the business categorize its customers into separate groups on the basis of characteristics and to send targeted emails and offers.
- Personalised Campaigns, Fraud Detection, Document Segmentation

## CHOICE MODEL

### What it is?

The choice model links to the second market principle that consumers' preferences change. It is a scientific approach employed by academics, economists, and policy analysts to measure preference. It is considered the least biased scientific method to study and understand how choices are made.

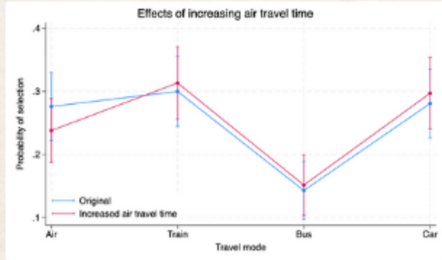
It is used to predict consumer choice in order to predict shifts in consumer value and use.

### how it works?

- Choice modelling relies on data collection, including various attributes that are represented in mathematical models that stimulate behavior. The objective is to be able to predict the likelihood of an individual choosing a specific option.
- The Multinomial Logit (MNL) is a commonly applied choice model.

$$P_i = \frac{e^{U_i}}{\sum_{j=1}^{n} e^{U_j}}$$

Pi = probability of choosing option i
Ui = utility of option i
n = total number of alternatives




*The graph shows the impact that an increase in air travel time has on the probability of choosing air travel, while slightly increasing the probability of a mode choice to train, bus, and car.*

Source: StataCorp (n.d.). Choice models. stata.com

### Managerial Implications

- Assess consumer price sensitivity.
- Estimate volumes and market share.
- Estimate the adoption of new products or services.

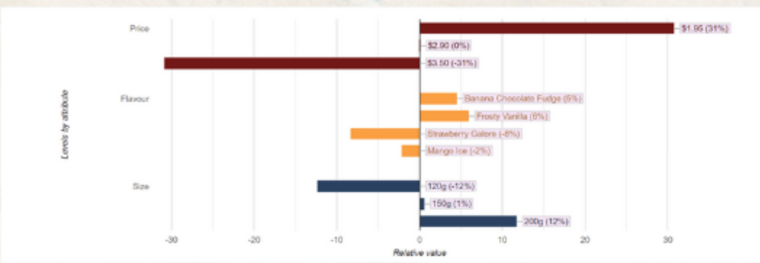
## CONJOINT ANALYSIS

### What it is?

Conjoint analysis is a very effective method for market research because it measures how consumers prefer a product/service based on different attributes (features, price, brand). Conjoint analysis is based on the notion that the value a consumer places on an offering is the sum of its parts, helping firms understand the trade-offs that customers make and getting part-worth utilities for each attribute.

### how it works?

The task is to classify the customers who will respond to the next marketing campaign on the basis of information collected about the customers and into two classes of respondents and nonrespondents.




*This graph demonstrates that larger and less expensive cones were preferred, and the second two most popular flavours were fudge and vanilla, respectively.*

Source: Conjoint.ly (2017). What is Conjoint Analysis?. Conjointly.com.

### Managerial Implications

- The insights derived from conjoint analysis of a company's product features can be used in a variety of ways. More often than not, conjoint analysis can affect pricing strategy, sales and marketing efforts, and research and development planning.

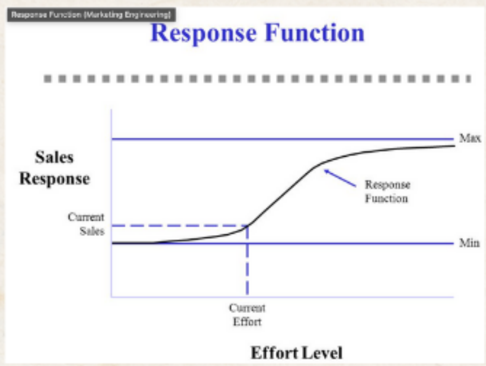
## MARKET RESPONSE MODELS

### What it is?

Market response models seek to help researchers and practitioners understand and quantify how Individual and collective consumers respond to marketing activity, as well as how competitors interact with each other. The appropriate estimated effects are a starting point for better decisions making in marketing

### how it works?

These models often use regression to establish a cause-and-effect relationship between marketing inputs (e.g., advertising spend, pricing, distribution, and promotion) and sales outputs.




*The graph shows how sales respond to different levels of marketing effort, which typically has diminishing returns at higher investment levels.*

Source:Nguyen (n.d.). Marketing response model. bookdown.org

### Managerial Implications

- The insights derived from conjoint analysis of a company's product features can be used in a variety of ways. More often than not, conjoint analysis can affect pricing strategy, sales and marketing efforts, and research and development planning.

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